

Section 19 - Composition API

What is the "Composition API"?

But you **might face two main limitations / issues** when building bigger Vue apps.

Code that **belongs together logically** is **split up** across multiple options (data, methods, computed)

Re-using logic across components can be **tricky or cumbersome**

```
{  
  data() {  
    return { name: 'Max' };  
  },  
  methods: { doSth() { ... } }  
}
```



Composition API

```
{  
  setup() {  
    const name = ref('Max');  
    function doSth() { ... }  
    return { name, doSth };  
  }  
}
```

Core Features in Composition API

1 It seems like Composition API has made Vue just like React.

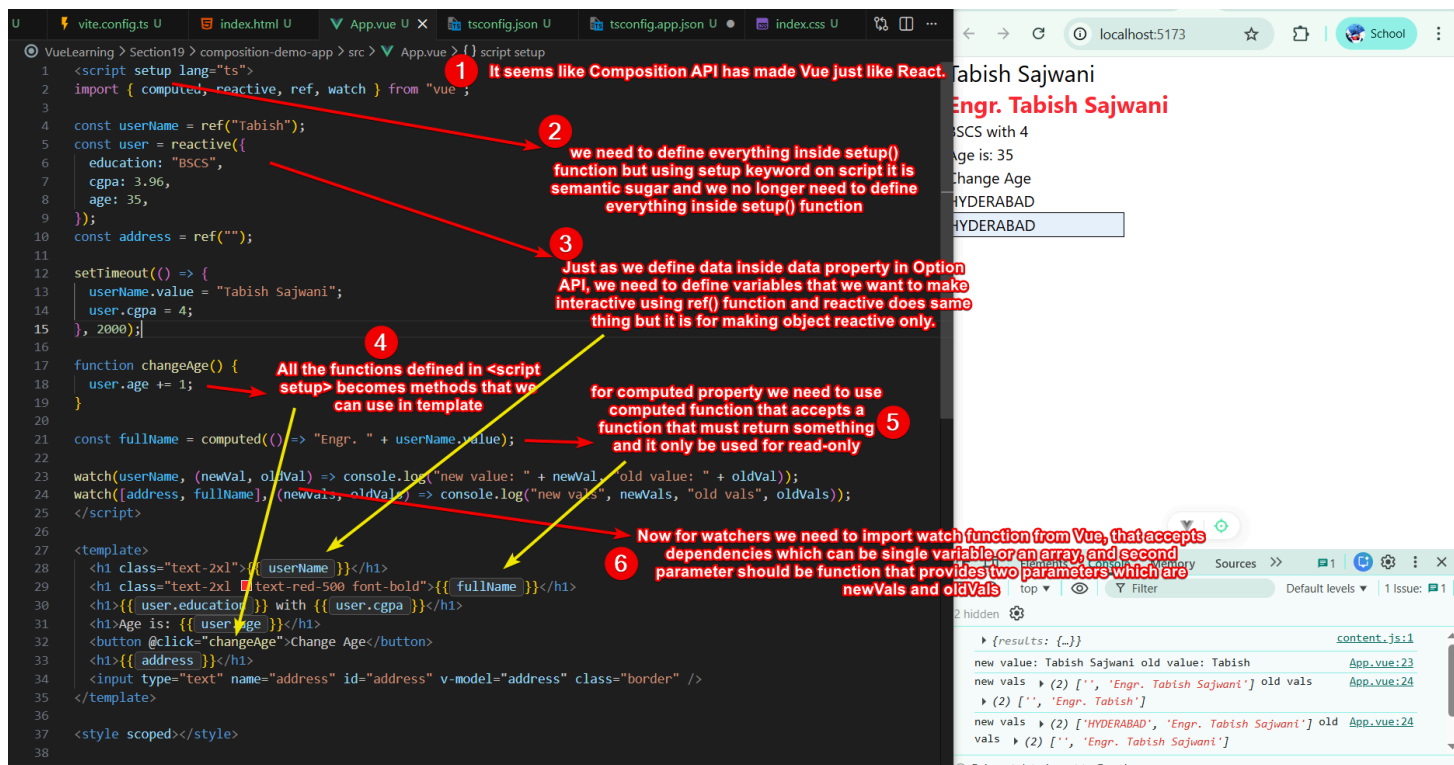
2 we need to define everything inside setup() function but using setup keyword on script it is semantic sugar and we no longer need to define everything inside setup() function

3 Just as we define data inside data property in Option API, we need to define variables that we want to make interactive using ref() function and reactive does same thing but it is for making object reactive only.

4 All the functions defined in <script setup> becomes methods that we can use in template

5 for computed property we need to use computed function that accepts a function that must return something and it only be used for read-only

6 Now for watchers we need to import watch function from Vue, that accepts dependencies which can be single variable or an array, and second parameter should be function that provides two parameters which are newVals and oldVals



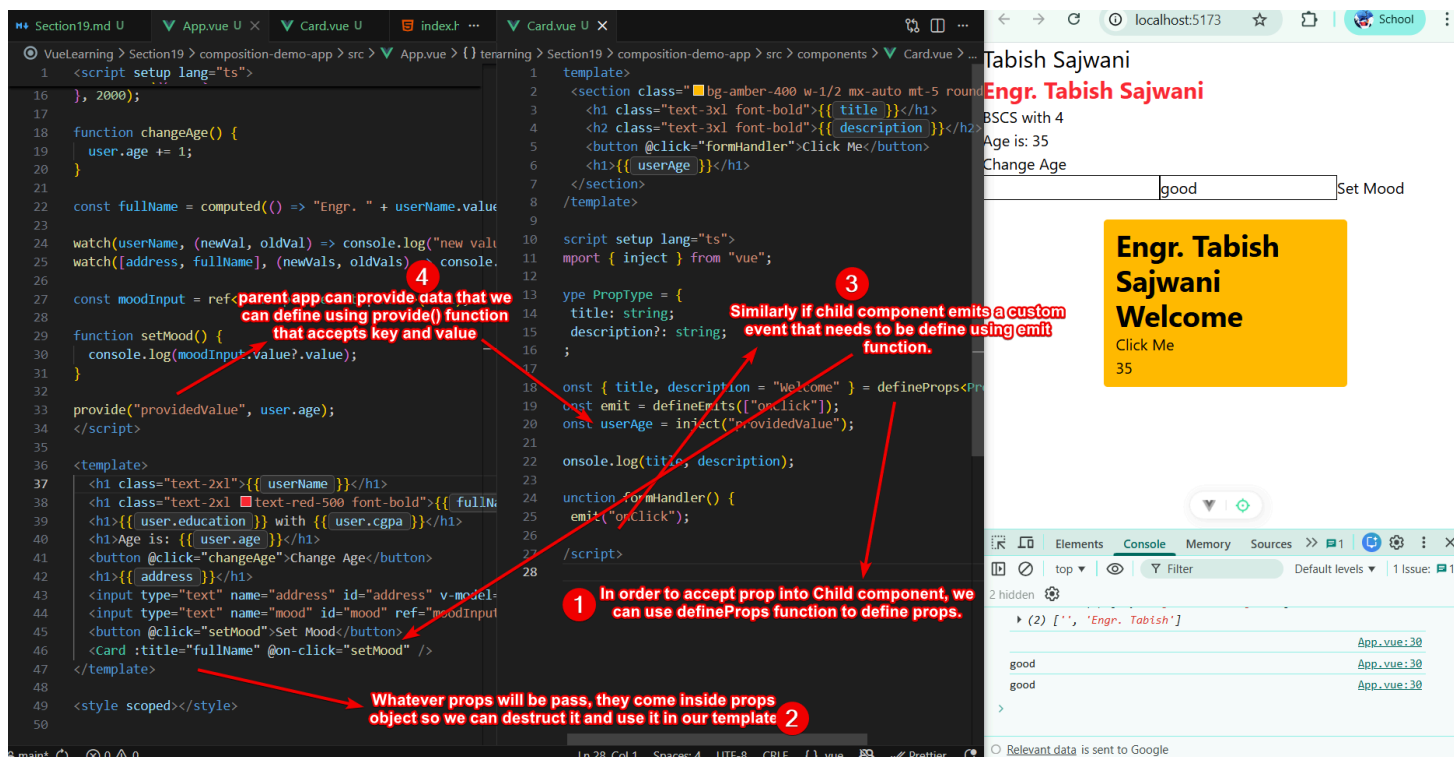
Props, Emit, and Provide/Inject

1 parent app can provide data that we can define using provide() function that accepts key and value

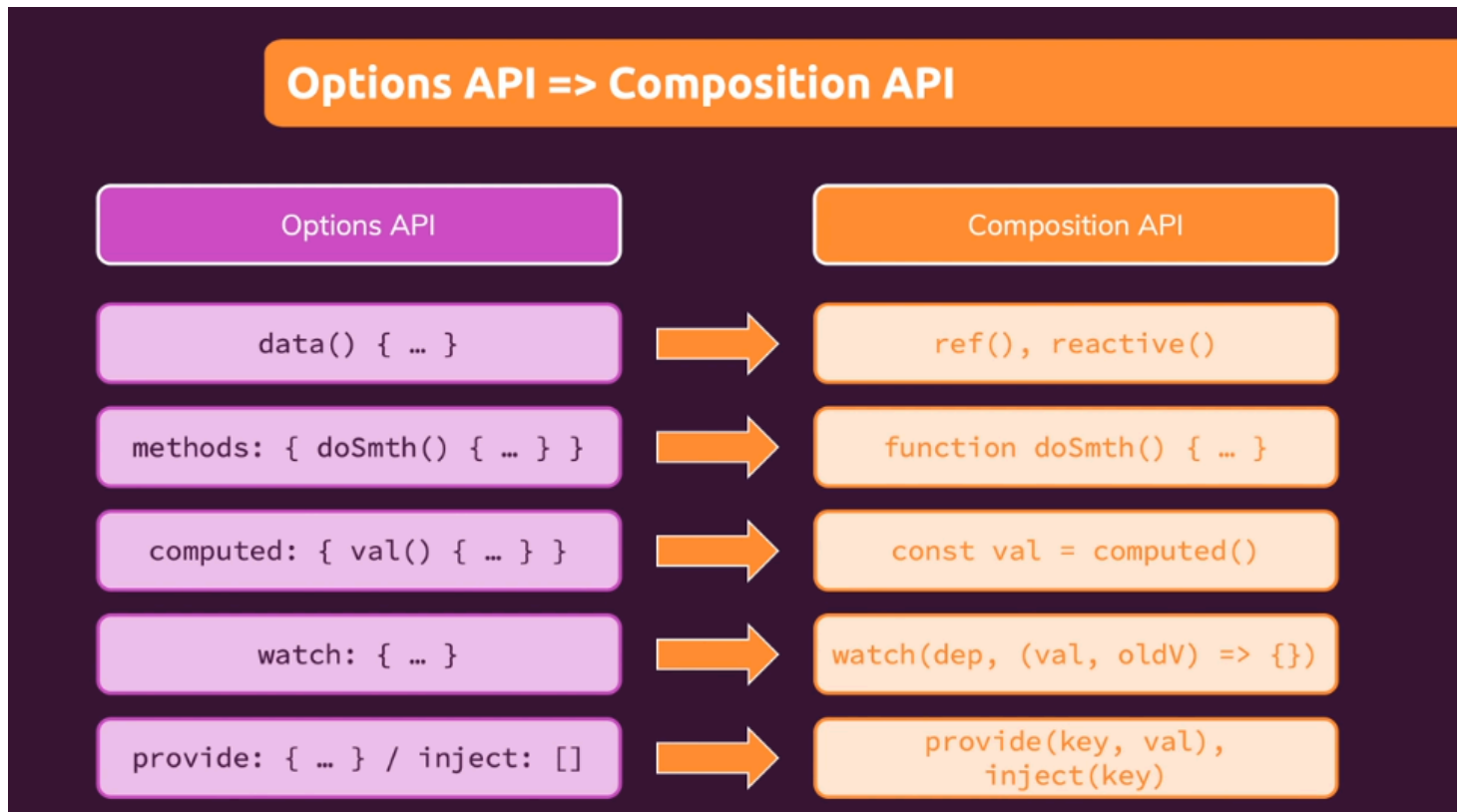
2 Similarly if child component emits a custom event that needs to be define using emit function.

3 In order to accept prop into Child component, we can use defineProps function to define props.

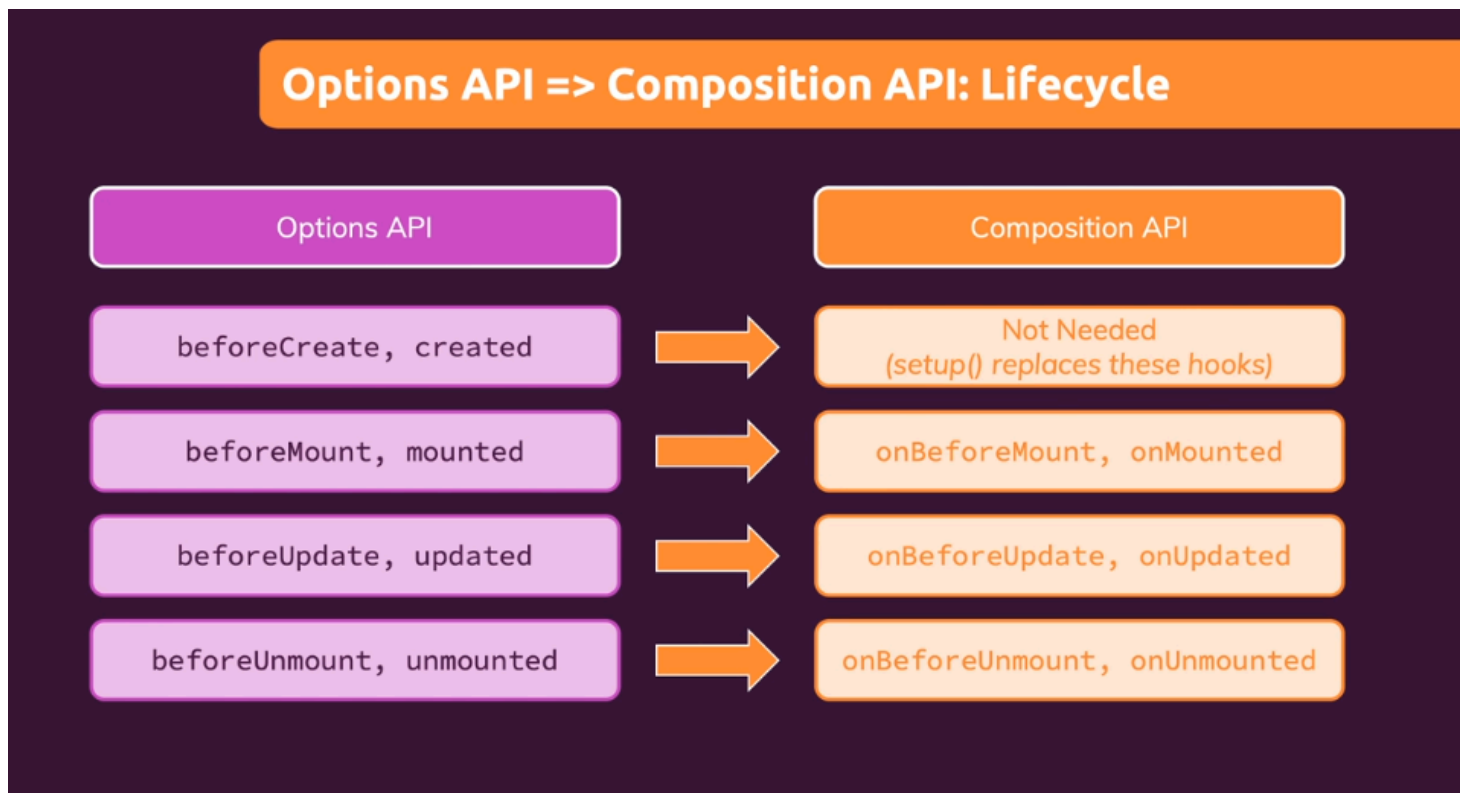
4 Whatever props will be pass, they come inside props object so we can destruct it and use it in our template



Option API to Composition API Summary



Lifecycle Hooks



Composition API Summary

Composition API

What & Why

It's an **alternative** to the Options API:
It uses **setup()** to **expose** logic/ data to the template

It's a **function-based solution** that allows you to keep **logically related code together**

Methods, Computed, Watchers

Methods become **regular functions** defined in setup()

Computed properties and **watchers** are **defined with imported functions** (from vue)

Data & Reactivity

Data can be managed as **ref()**s (individual values or objects) or **reactive()** objects

Reactivity is a key concept – refs and reactive objects are **reactive**, their **nested values are not**

The setup() Function

The setup() function is called by Vue when the **component is created** – it **defines data + logic** for the template

setup() receives **two arguments** (automatically): **reactive props** and **context** (attrs, slots, emit())